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Marlow

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(54) **LOW-POWER SOURCE OF SQUEEZED LIGHT**

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G02F 1/35 (2006.01)

(Continued)

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See application file for complete search history.

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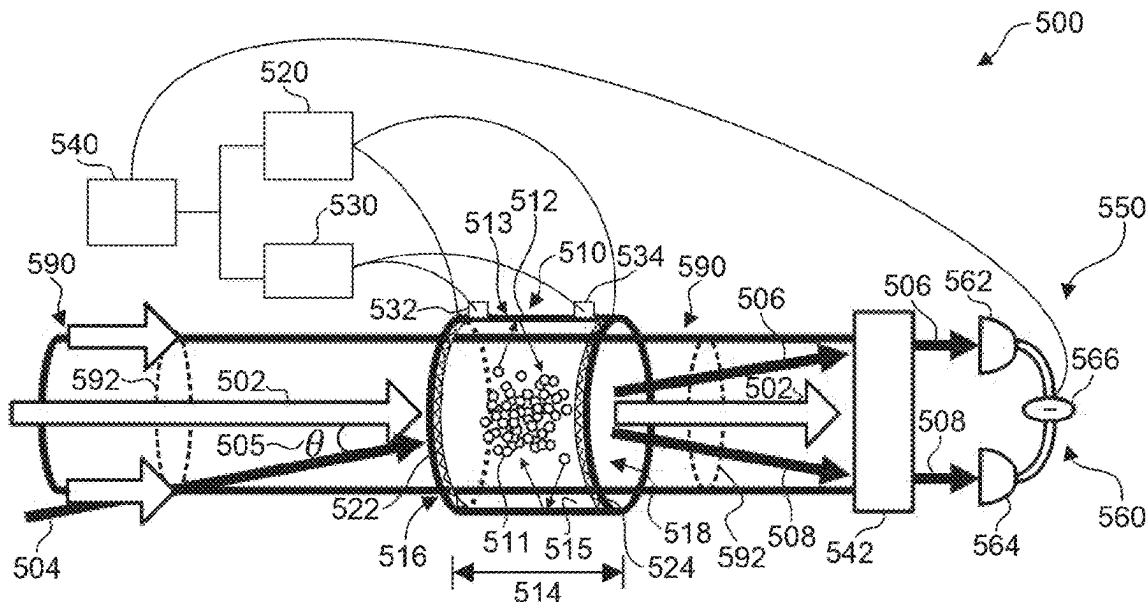
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(57) **ABSTRACT**

A degenerate four-wave mixing (DFWM) squeezed light apparatus includes one or more pump beams, a probe beam, a vapor cell, a repump beam, and a detector. The one or more pump beams includes an input power of no greater than about 150 mW. The vapor cell includes an atomic vapor configured to interact with overlapped pump and probe beams to generate an amplified probe beam and a conjugate beam. The repump beam is configured to optically pump the atomic vapor to a ground state and decrease atomic decoherence of the atomic vapor. The detector is configured to measure squeezing due to quantum correlations between the amplified probe beam and the conjugate beam. The one or more pump beams, the probe beam, and the repump beam are configured to generate two-mode squeezed light by DFWM with squeezing of at least 3 dB below shot noise.

20 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/117,331, filed on Dec. 10, 2020, now Pat. No. 11,637,408.

- (60) Provisional application No. 62/946,539, filed on Dec. 11, 2019.
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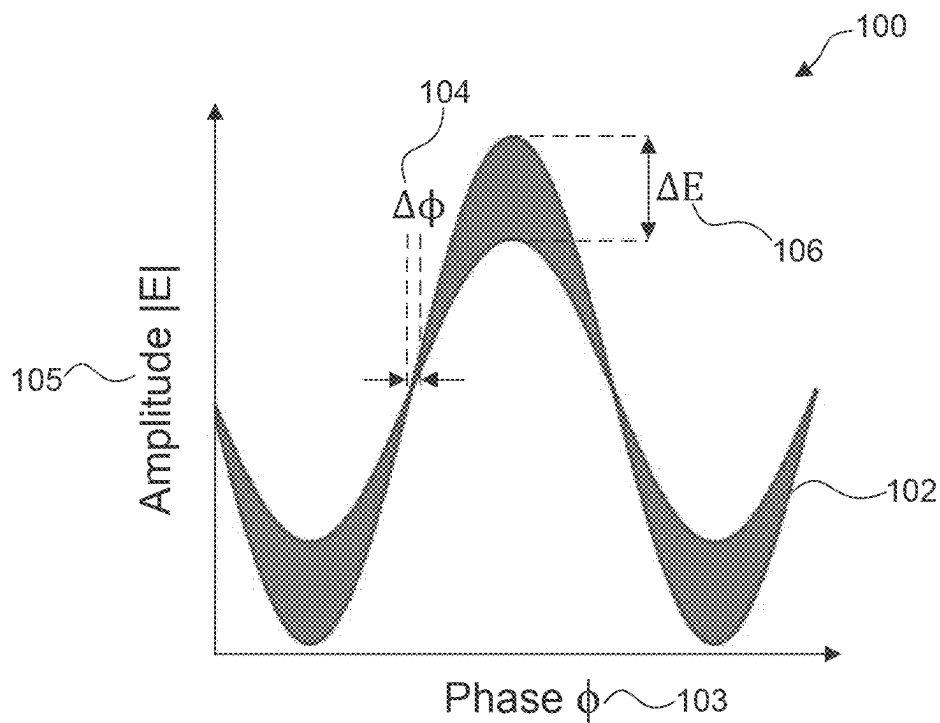


FIG. 1

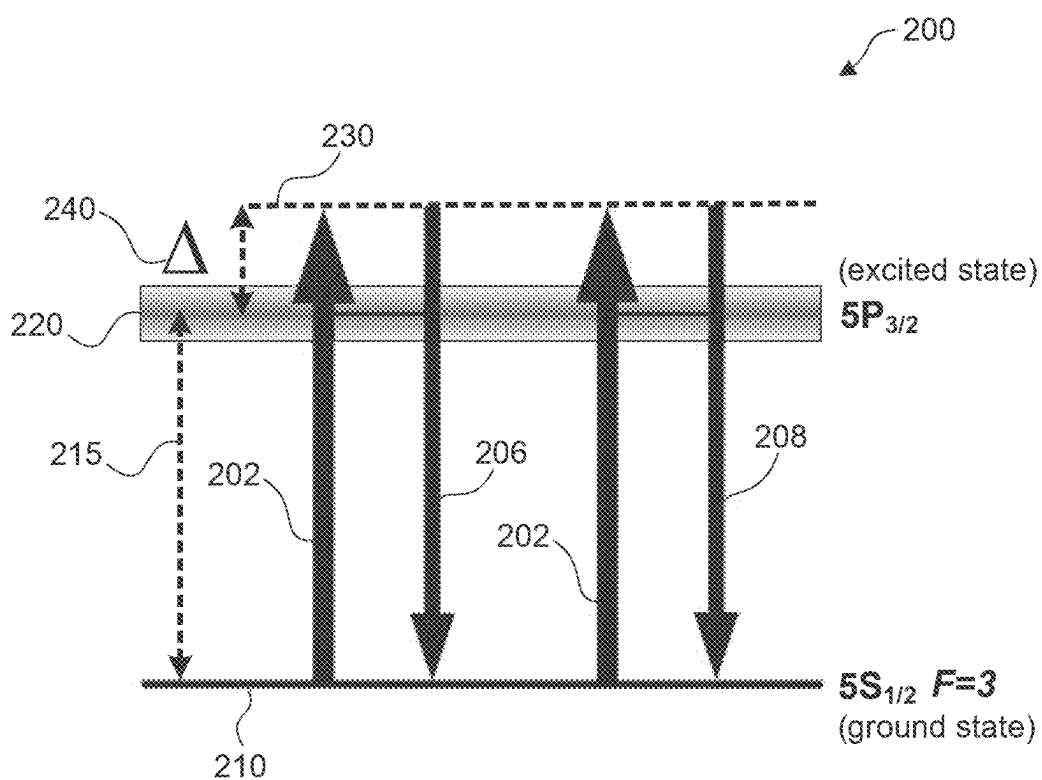


FIG. 2

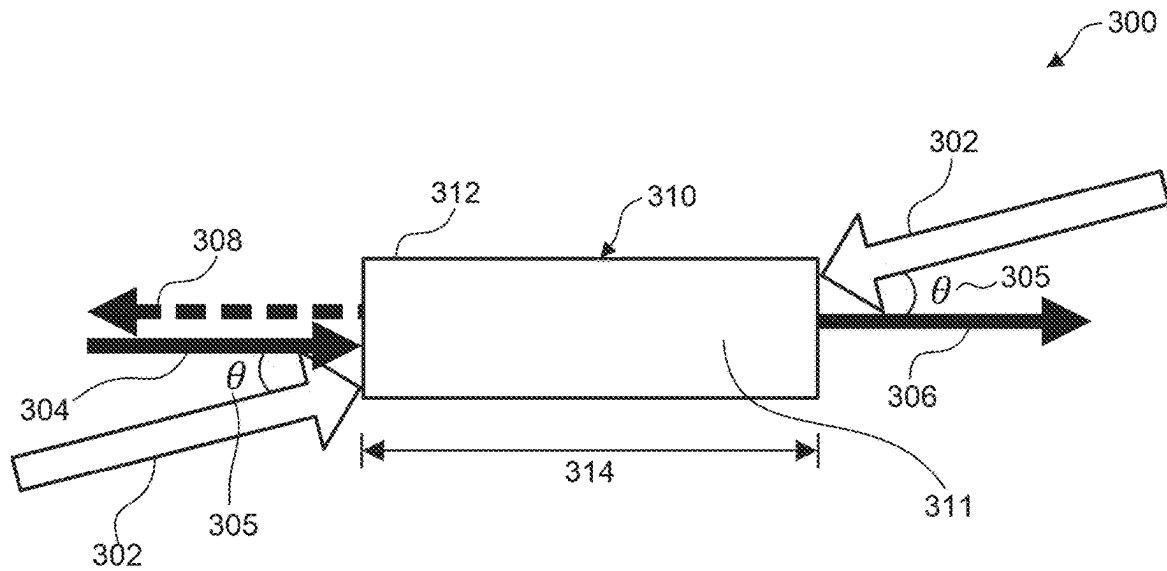


FIG. 3

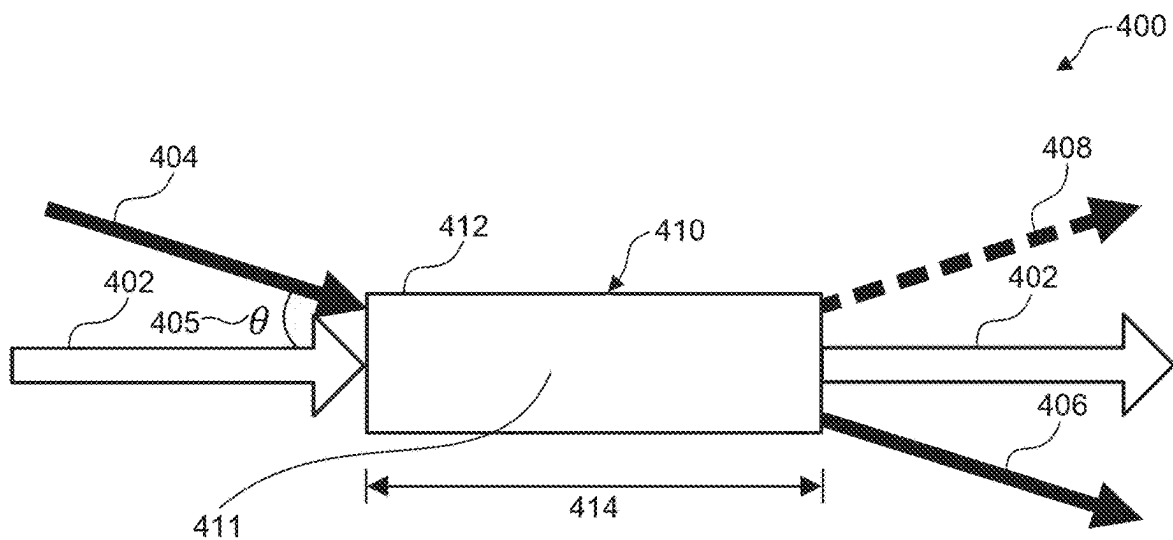


FIG. 4

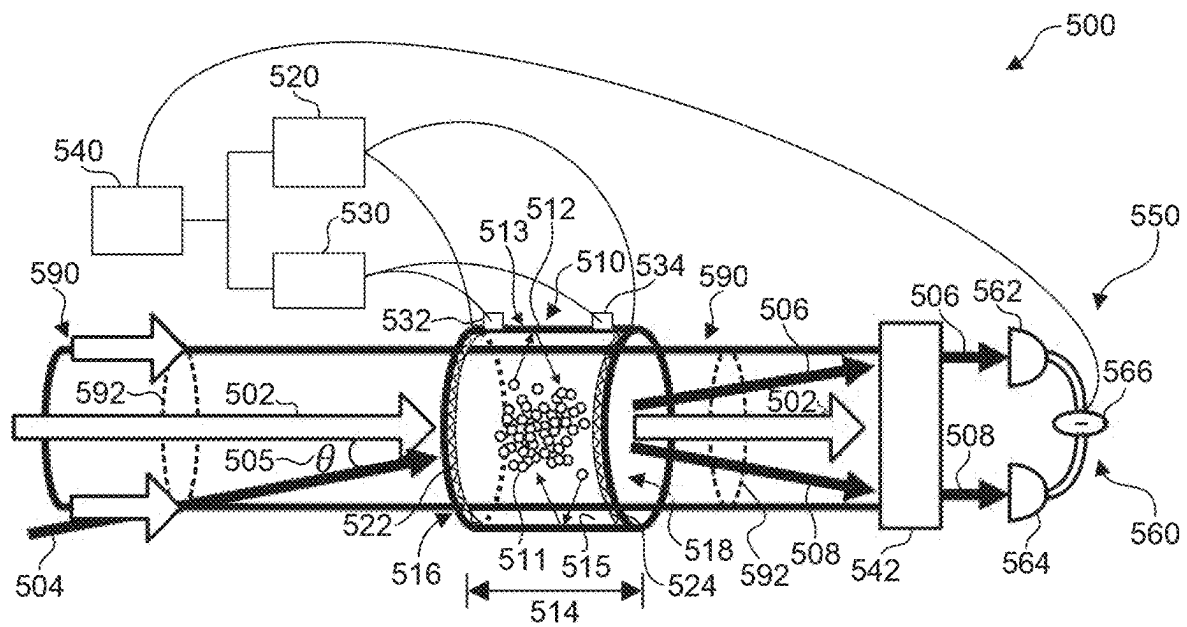


FIG. 5

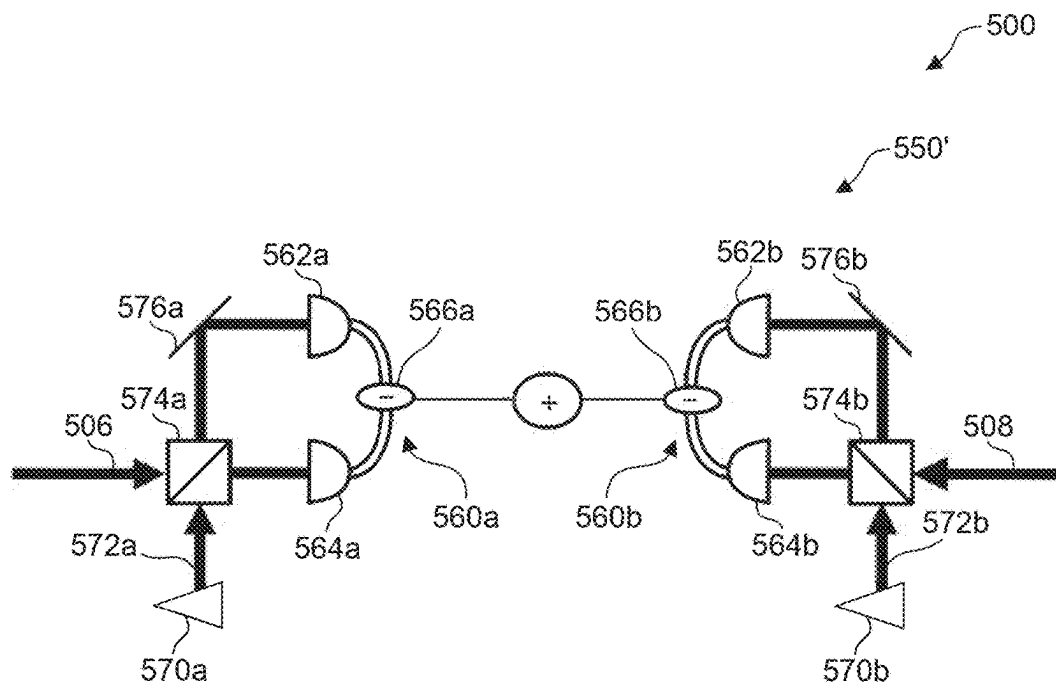


FIG. 6

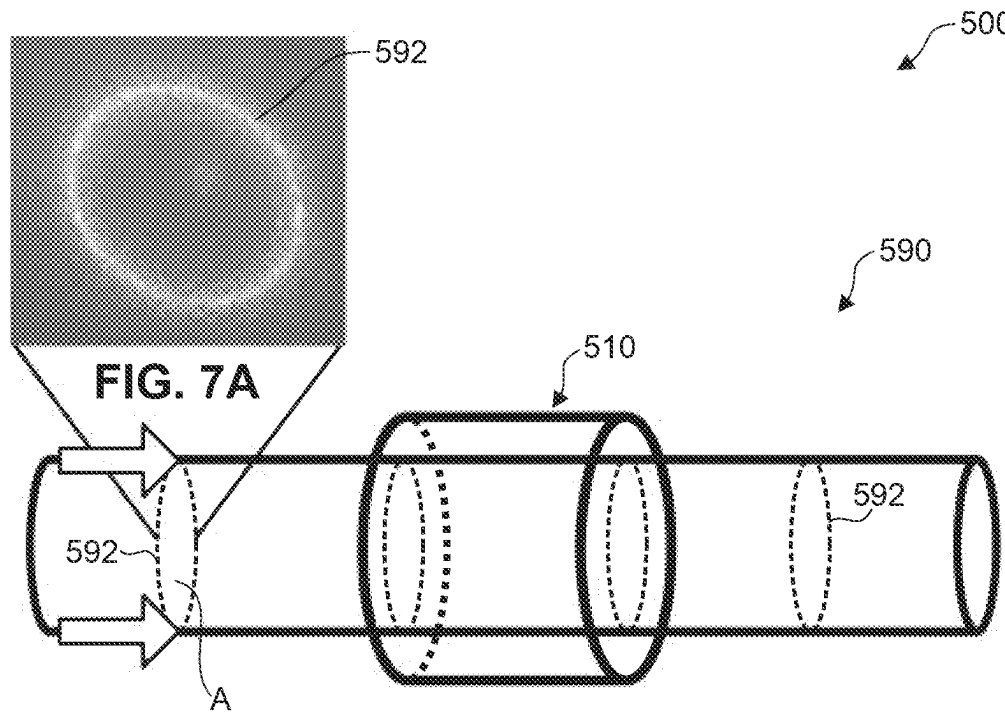


FIG. 7

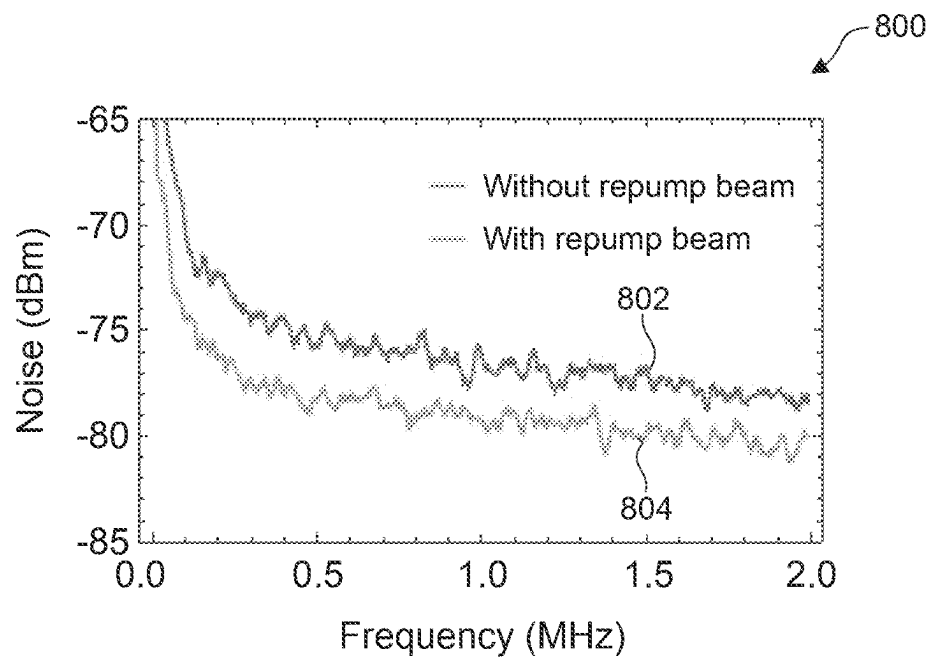


FIG. 8

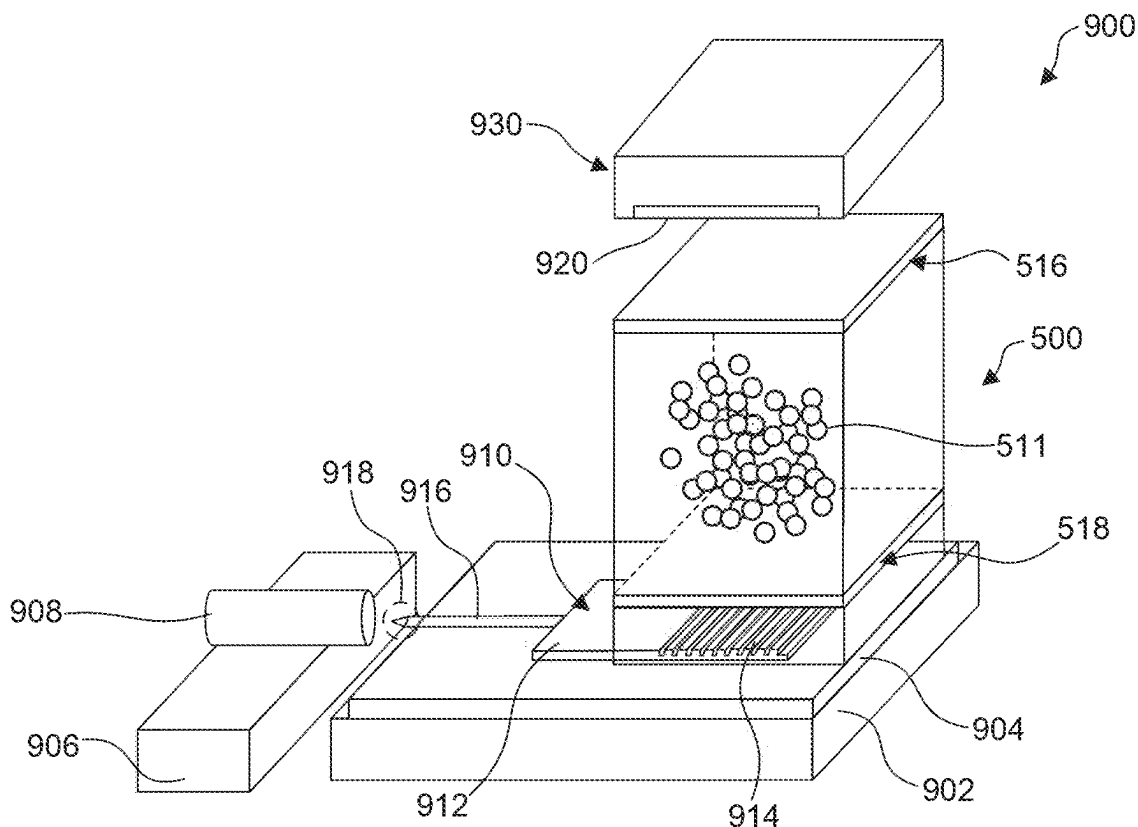


FIG. 9

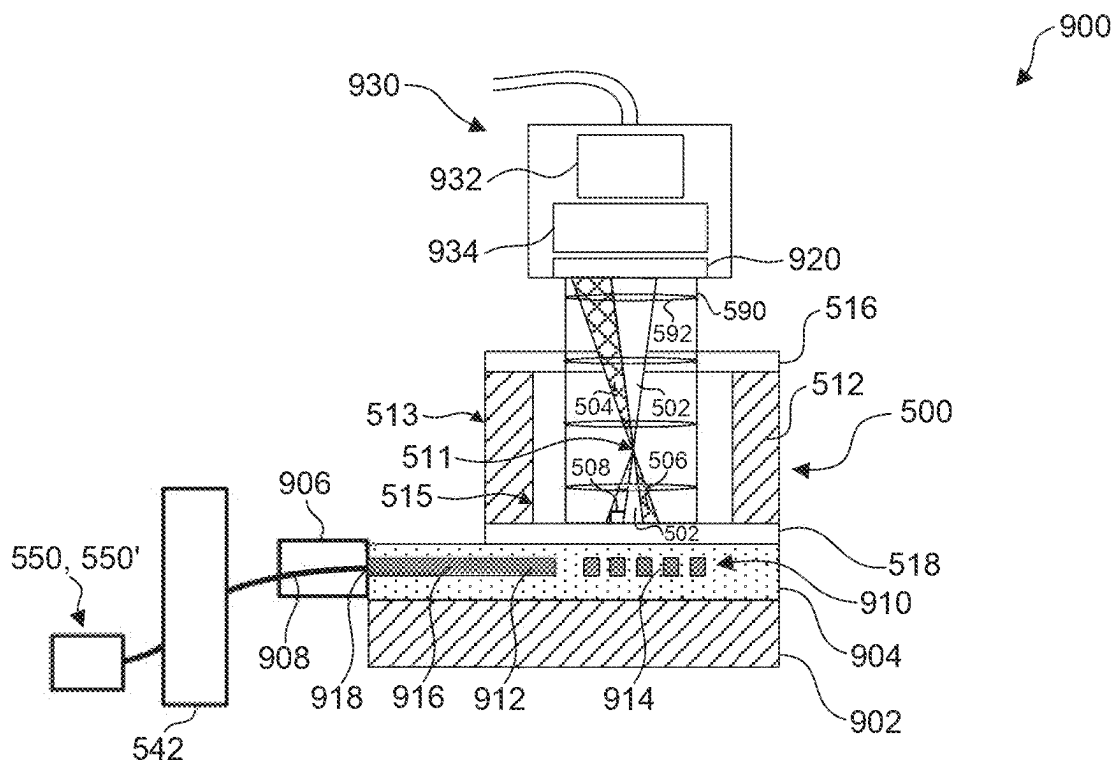


FIG. 10

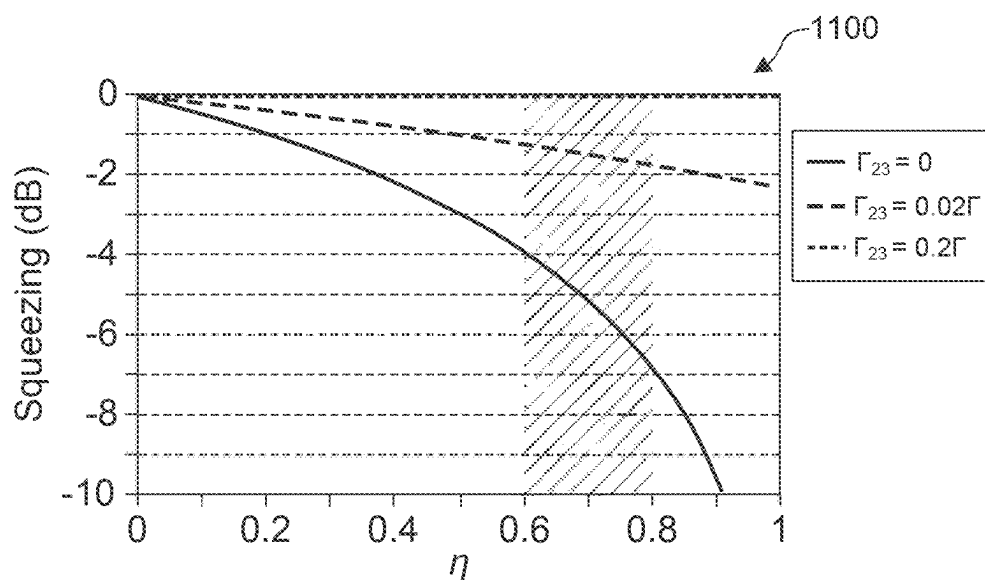


FIG. 11

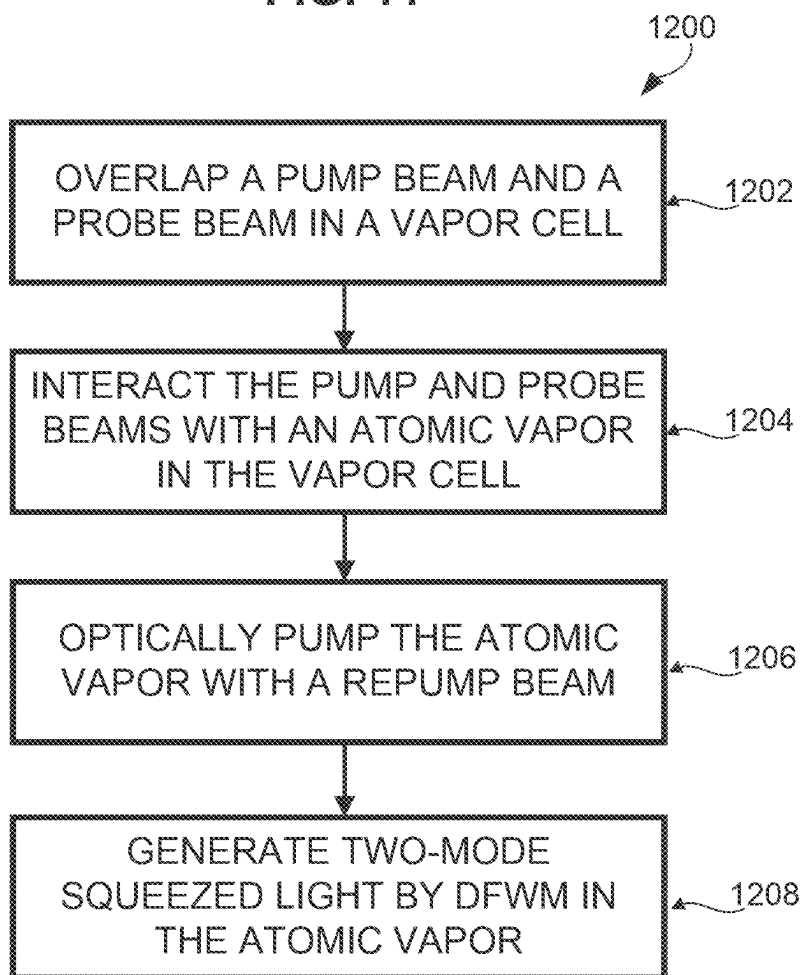


FIG. 12

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LOW-POWER SOURCE OF SQUEEZED LIGHT**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 18/121,186, filed Mar. 14, 2023, which is a continuation of U.S. application Ser. No. 17/117,331, filed Dec. 10, 2020, now U.S. Pat. No. 11,637,408, which claims priority to U.S. Provisional Application No. 62/946,539, filed Dec. 11, 2019, which are hereby incorporated herein in their entireties by reference.

BACKGROUND**Field**

The present disclosure generally relates to squeezed light apparatuses and systems, for example, low-power squeezed light sources.

Background

In quantum physics, an oscillating physical quantity (e.g., a light wave) cannot have precisely defined values at all phases of the oscillation. Squeezed light is a type of non-classical light in which one of its field quadratures has a reduced or “squeezed” noise at the expense of added noise in the other quadrature. Squeezed light can have a quantum noise reduction below a shot noise level, which has applications in precision optical measurements and quantum communication. Shot noise is the noise level that would be measured by coherent light of the same optical power when using the same detection scheme. Current techniques to generate quadrature-squeezed light require various optical components and high input powers and, thus, are impractical for scaling to a low-power and portable device.

SUMMARY

Accordingly, there is a need to provide a low-power and portable squeezed light source with a reduced size, weight, and power (SWaP) to improve the precision of optical measurements and implement continuous-variable quantum communication protocols. To meet this need using degenerate four-wave mixing (DFWM), excess noise must be reduced.

In some embodiments, a degenerate four-wave mixing (DFWM) squeezed light apparatus includes one or more pump beams, a probe beam, a vapor cell, a repump beam, and a balanced differential detector or a joint homodyne detector. The one or more pump beams includes an input power of no greater than about 150 mW. The probe beam is configured to overlap the one or more pump beams. The vapor cell includes an atomic vapor, a first window, and a second window. The atomic vapor is configured to interact with the overlapped pump and probe beams to generate an amplified probe beam and a conjugate beam. The repump beam is configured to optically pump the atomic vapor to a ground state and decrease atomic decoherence of the atomic vapor. The balanced differential detector or the joint homodyne detector is configured to measure squeezing due to quantum correlations between the amplified probe beam and the conjugate beam. The one or more pump beams, the probe beam, and the repump beam are configured to reduce spontaneous emission of the atomic vapor and generate

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two-mode squeezed light by DFWM in the atomic vapor with squeezing of at least 3 dB below shot noise.

In some embodiments, the input power of the one or more pump beams is no greater than about 50 mW. In some embodiments, the input power of the one or more pump beams is no greater than about 20 mW.

In some embodiments, the repump beam is cylindrical. In some embodiments, the repump beam is an annulus. In some embodiments, the repump beam includes two counter-propagating annular repump beams. In some embodiments, an input power of the repump beam is no greater than about the input power of the one or more pump beams.

In some embodiments, the one or more pump beams, the probe beam, and the vapor cell are arranged in a forward-scattering geometry. In some embodiments, the one or more pump beams, the probe beam, and the vapor cell are arranged in a backward-scattering geometry.

In some embodiments, the two-mode DFWM squeezed light includes squeezing of at least 6 dB below shot noise. In some embodiments, the balanced differential detector includes an intensity-difference detector. In some embodiments, the joint homodyne detector includes a pair of homodyne detectors.

In some embodiments, the first and second windows have a temperature greater than an exterior surface of a cylindrical wall of the vapor cell. In some embodiments, a temperature of the vapor cell is about 30° C. to about 100° C. In some embodiments, the temperature is about 35° C. to about 45° C. In some embodiments, a longitudinal length of the vapor cell is about 0.5 cm to about 10 cm. In some embodiments, the atomic vapor includes a rubidium vapor.

In some embodiments, the DFWM squeezed light apparatus includes a portable diode laser. In some embodiments, the portable diode laser includes a volume of no greater than about 100 cm³. In some embodiments, the portable diode laser includes an output power of no greater than about 150 mW. In some embodiments, the DFWM squeezed light apparatus includes a portable integrated photonic chip.

In some embodiments, a method of forming a low-power squeezed light source includes overlapping one or more pump beams and a probe beam. The one or more pump beams includes an input power of no greater than about 150 mW. In some embodiments, the method further includes interacting an atomic vapor with the overlapped pump and probe beams to generate an amplified probe beam and a conjugate beam. In some embodiments, the method further includes optically pumping the atomic vapor with a repump beam configured to decrease atomic decoherence of the atomic vapor. In some embodiments, the method further includes generating two-mode squeezed light by degenerate four-wave mixing (DFWM) in the atomic vapor with squeezing of at least 3 dB below shot noise.

In some embodiments, the overlapping includes applying an input power of the one or more pump beams no greater than about 50 mW. In some embodiments, the optical pumping includes an annular repump beam. In some embodiments, the two-mode DFWM squeezed light includes squeezing of at least 6 dB below shot noise. In some embodiments, the method further includes measuring an intensity-difference squeezing due to quantum correlations between the amplified probe beam and the conjugate beam with a balanced differential detector. In some embodiments, the method further includes measuring quadrature squeezing due to quantum correlations between the amplified probe beam and the conjugate beam with a joint homodyne detector.

Further features and exemplary aspects of the embodiments, as well as the structure and operation of various embodiments, are described in detail below with reference to the accompanying drawings. It is noted that the embodiments are not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the embodiments and, together with the description, further serve to explain the principles of the embodiments and to enable a person skilled in the relevant art(s) to make and use the embodiments.

FIG. 1 is a schematic illustration of a squeezed light wave, according to some embodiments.

FIG. 2 is a schematic illustration of an energy diagram for degenerate four-wave mixing (DFWM) in a two-level atomic system, according to some embodiments.

FIG. 3 is a schematic illustration of a backward-scattering geometry, according to some embodiments.

FIG. 4 is a schematic illustration of a forward-scattering geometry, according to some embodiments.

FIG. 5 is a schematic perspective illustration of a DFWM squeezed light apparatus, according to some embodiments.

FIG. 6 is a schematic illustration of a joint homodyne detector, according to some embodiments.

FIG. 7 is a schematic perspective illustration of the repump beam of the DFWM squeezed light apparatus of FIG. 5, according to some embodiments.

FIG. 7A is a schematic partial cross-sectional illustration of the repump beam of FIG. 7, according to some embodiments.

FIG. 8 is a schematic illustration of a plot of noise, according to some embodiments.

FIG. 9 is a schematic perspective illustration of a DFWM squeezed light system, according to some embodiments.

FIG. 10 is a schematic cross-sectional illustration of the DFWM squeezed light system of FIG. 9, according to some embodiments.

FIG. 11 is a schematic illustration of a plot of squeezing, according to some embodiments.

FIG. 12 illustrates a flow diagram for forming a low-power squeezed light source, according to some embodiments.

The features and exemplary aspects of the embodiments will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears. Unless otherwise indicated, the drawings provided throughout the disclosure should not be interpreted as to-scale drawings.

DETAILED DESCRIPTION

The embodiment(s) described, and references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment(s) described may include a particular

feature, structure, and/or characteristic, but every embodiment may not necessarily include the particular feature, structure, and/or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, and/or characteristic is described in connection with an embodiment, it is understood that it is within the knowledge of one skilled in the art(s) to effect such feature, structure, and/or characteristic in connection with other embodiments whether or not explicitly described. The scope of this disclosure is not limited to the disclosed embodiment(s) but is instead defined by the claims appended hereto.

Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “on,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or in operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

The term “about” or “substantially” or “approximately” as used herein indicates the value of a given quantity that can vary based on a particular technology. Based on the particular technology, the term “about” or “substantially” or “approximately” can indicate a value of a given quantity that varies within, for example, 1-15% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 5\%$, $\pm 10\%$, or $\pm 15\%$ of the value).

Embodiments of the disclosure may be implemented in hardware, firmware, software, or any combination thereof. Embodiments of the disclosure may also be implemented as instructions stored on a machine-readable medium, which may be read and executed by one or more processors. A machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computing device). For example, a machine-readable medium may include read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory devices, and/or electrical, optical, acoustic, or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.), and others. Further, firmware, software, routines, and/or instructions may be described herein as performing certain actions. However, it should be appreciated that such descriptions are merely for convenience and that such actions in fact result from computing devices, processors, controllers, and/or other devices executing the firmware, software, routines, instructions, etc.

Exemplary DFWM Squeezed Light Apparatuses

As discussed above, an oscillating physical quantity (e.g., a light wave) cannot have precisely defined values at all phases of the oscillation. In quantum optics, an optical phase space is a phase space in which all quantum states of an optical system are described. Each point in the optical phase space corresponds to a unique state (e.g., $|\alpha\rangle$) of the optical system.

Under the Heisenberg uncertainty principle, quantum uncertainty exists for certain pairs of physical properties. For example, quantum uncertainty exists for a light wave's quadratures, the electric field phase (ϕ) and the electric field amplitude ($|E|$). Physical properties, such as the electric field phase (ϕ) and the electric field amplitude ($|E|$), can become quantum operators. In quantum physics, an operator is a

function over a space of physical states onto another space of physical states. Quadratures are defined as operators (e.g., X_1 and X_2) that represent the real and imaginary parts of a complex state a (e.g., the in-phase and the out-of-phase components of the light wave). The light wave can be represented by $E(t)=E_0[X_1 \sin(\omega t+\phi)-X_2 \cos(\omega t+\phi)]$, where X_1 =amplitude quadrature and X_2 =phase quadrature. The measured electric field amplitude ($|E|$) at the light wave's electric field phase (ϕ) are the eigenvalues (e.g., scalar results of a linear transformation) of the normalized quadrature operator X_ϕ , where $X_{\phi=0}=X_1$ is the light wave's amplitude quadrature and $X_{\phi=90}=X_2$ is the light wave's phase quadrature. X_1 and X_2 are non-commuting observables (e.g., no joint quantum probability, $[X_1, X_2] \neq 0$) and satisfy $\Delta X_1 \Delta X_2 \geq 1/4$.

Quantum uncertainty is visible when identical measurements of the same quantity (e.g., observable) on identical objects (e.g., modes of light) give different results (e.g., eigenvalues). In quantum physics, a light wave is in a squeezed state if its electric field amplitude ($|E|$) for some phase (ϕ) has a quantum uncertainty less than that of a coherent state (e.g., a state of the quantized electric field with maximal coherence and classical behavior). Thus, squeezing refers to a reduced quantum uncertainty. Further, a squeezed state must also have some phase (ϕ) at which the electric field amplitude ($|E|$) is anti-squeezed (e.g., larger than a coherent state), in order to obey the Heisenberg uncertainty principle.

A squeezed state is a quantum state described by two non-commuting observables (e.g., $[X_1, X_2] \neq 0$) having a continuous spectra of eigenvalues (e.g., unbounded). The continuous spectra of eigenvalues, as opposed to a discrete spectra (e.g., bound states), provides for continuous-variable states (e.g., continuous value of energies) that can be used for universal quantum communication. For example, a squeezed state can be described by the light wave's electric field in the amplitude quadrature X_1 (e.g., phase 0°) and in the phase quadrature X_2 (e.g., phase 90°). Under the uncertainty principle, the product of the standard deviations of the amplitude X_1 and phase X_2 quadratures is $\Delta X_1 \Delta X_2 \geq 1/4$. A quadrature-squeezed state of light is characterized by noise whose standard deviation in one quadrature is below that of coherent light of the same optical power.

Squeezed light is a type of non-classical light in which one of its field quadratures (e.g., amplitude X_1 or phase X_2) has a reduced ("squeezed") noise at the expense of added noise in the other quadrature. Squeezed light has a reduced quantum uncertainty while anti-squeezed light has a larger quantum uncertainty. Diminishing the quantum noise at a specific quadrature (e.g., phase X_2) of a light wave increases the noise of the complementary quadrature (e.g., amplitude X_1).

FIG. 1 illustrates squeezed light wave **100**, according to various exemplary embodiments. Squeezed light wave **100** can include an oscillating electric field **102** (e.g., $E(t)$) with a first quadrature (e.g., phase **103**) and a second quadrature (e.g., amplitude **105**). As shown in FIG. 1, oscillating electric field **102** can include an anti-squeezed or increased amplitude uncertainty **106** and a squeezed or reduced phase uncertainty **104** (e.g., phase-squeezed light). Reduced phase uncertainty **104** can have a reduced quantum noise compared to the phase noise of a coherent field of the same optical power (e.g., shot noise). In some embodiments, oscillating electric field **102** can include an anti-squeezed or increased phase uncertainty and a squeezed or reduced amplitude uncertainty (e.g., amplitude-squeezed light).

Two-mode squeezing involves two modes of the electric field which exhibit quantum noise reduction below the shot noise level in a linear combination of the quadratures of the fields (e.g., phase and amplitude). Shot noise is the noise level that would be measured by coherent light of the same optical power when using the same detection scheme. Quantum information processing can be carried out with continuous variables in a parallel fashion to the ways in which binary data is processed. Physical quantities with a continuous spectrum (e.g., phase and amplitude) can be used for continuous-variable entanglement. Two-mode squeezing can be exploited to generate continuous-variable entanglement.

Continuous-variable entanglement refers to light beams in Einstein-Podolsky-Rosen (EPR) or entangled states that can be used for quantum information protocols. A continuous-variable system is a canonical infinite dimensional quantum system comprised of N bosonic (e.g., photonic) modes with an associated Hilbert space $H=\bigotimes_{k=1}^N H_k$. A Hilbert space is an abstract vector space possessing the structure of an inner product that allows length and angle to be measured. Each of these modes H_k has an associated annihilation and creation operator $\hat{a}$, $\hat{a}^\dagger$, respectively. These operators obey the commutation relations $[\hat{a}_i, \hat{a}_j^\dagger]=\delta_{ij}$ and $[\hat{a}_i, \hat{a}_j]=[\hat{a}_i^\dagger, \hat{a}_j^\dagger]=0$. The space H_k is spanned by the Fock basis $\{|n_k\rangle\}_{n=0}^\infty$ of eigenstates of the number operator $\hat{n}_k=\hat{a}_k^\dagger \hat{a}_k$. The Fock basis (space) constructs the quantum states space of a variable or unknown number of identical particles (e.g., photons) from a single particle Hilbert space (H_k). These eigenstates have the property that $\hat{n}|n\rangle=n|n\rangle$, $\hat{a}|n\rangle=\sqrt{n}|n-1\rangle$, and $\hat{a}^\dagger|n\rangle=\sqrt{n+1}|n+1\rangle$, as well as the fact that the vacuum state $|0\rangle$ is annihilated by $\hat{a}|0\rangle=0$. In the absence of any interactions, these modes evolve according to the Hamiltonian $H=\sum_{k=1}^N (\hat{a}_k^\dagger \hat{a}_k + 1/2)$. Two quadrature operators $\hat{q}_k=\hat{a}_k+\hat{a}_k^\dagger$ and $\hat{p}_k=i(\hat{a}_k^\dagger - \hat{a}_k)$ (e.g., amplitude quadrature X_1 and phase quadrature X_2) can be defined, which act in a similar fashion to the position and momentum operators in a quantum harmonic oscillator. Continuous-variable entanglement can be generated by multiple photons in one light mode and/or nonlinear optical processes (e.g., optical frequency comb, optical parametric oscillator, non-degenerate four-wave mixing, etc.).

Squeezed light can be generated using nonlinear optical processes (e.g., non-degenerate four-wave mixing, parametric down-conversion, etc.). However, these techniques require various optical components, high input powers (e.g., greater than 500 mW), have low power conversion efficiencies (e.g., less than 0.1%), and are not currently compact (e.g., not easily portable). Thus, current techniques are impractical for scaling to a low-power and portable squeezed light source.

Four-wave mixing (FWM) is a nonlinear interaction between light and matter that permits the transfer of energy among four modes of the electric field via their interaction with a nonlinear medium (e.g., an atomic vapor). FWM is an intermodulation phenomenon in nonlinear optics, whereby interactions between two or three wavelengths produce two or one new wavelengths. Intermodulation is the amplitude modulation of signals containing two or more different frequencies, caused by nonlinearities or time variance in the system. When three frequencies (f_1 , f_2 , and f_3) interact with a nonlinear medium, a fourth frequency (f_4) is formed by the scattering of the incident photons. Given three input frequencies (f_1 , f_2 , and f_3), the nonlinear system will output $\pm f_1 \pm f_2 \pm f_3$.

In degenerate four-wave mixing (DFWM), all four frequencies (f_1 , f_2 , f_3 , and f_4) are the same. DFWM occurs if

three fields (e.g., electric fields) of the same frequency interact with a nonlinear medium (e.g., an atomic vapor) and generate and/or amplify a fourth field of the same frequency. For example, three laser beams of the same frequency can interact in a nonlinear medium (e.g., atomic vapor) to produce a fourth degenerate beam. At least three of the four fields must have distinct wavevectors (e.g., k-vectors), defining directions of propagation of the fields, but all four fields have the same frequency in vacuum. Compared to non-degenerate nonlinear optical processes, DFWM can generate squeezed light via a nonlinear optical process that requires fewer optical components and has a high power conversion efficiency (e.g., greater than 10%) at low input pump beam powers (e.g., less than 150 mW). For example, in rubidium (Rb), the $5S_{1/2}$ to $5P_{3/2}$ atomic transition (e.g., 780.2 nm) is easily accessible with a simple diode laser system.

In atomic physics and spectroscopy, fine structure is the splitting of spectral lines due to the coupling between the orbital and spin angular momenta of the valence electron. For example, for the $5S_{1/2}$ ground state of Rb, this total momentum, defined in units of Planck's constant $\hbar$, is designated by the fine structure subscript $1/2$ (e.g., $5S_{1/2}$). Fine structure is the splitting of the main spectra lines of an atom (e.g., Rb) into two or more components, each representing a slightly different wavelength. Fine structure is a property of the atom. Fine structure is measured when an atom (e.g., Rb) emits light when transitioning from one energy state to another (e.g., the $5S_{1/2}$ to $5P_{3/2}$ atomic transition at 780.2 nm).

Hyperfine structure is the additional splitting of the fine structure spectral lines due to the interaction between the valence electron's momentum to the nuclear angular momentum. In hyperfine structure, line splitting is attributed to the fact that in addition to electron spin in an atom (e.g., fine structure), the atomic nucleus itself spins about its own axis. Hyperfine structure results from the interactions of the nucleus (or nuclei) with internally generated electric and magnetic fields. Energy states of the atom will be split into levels corresponding to slightly different energies. Each of these energy levels may be assigned a quantum number (n) and are called quantized levels. Hyperfine structure is a property of the atom. Hyperfine structure is measured when an atom (e.g., Rb) emits light when transitioning between these quantized energy levels. Hyperfine structure is characterized by energy shifts (e.g., wavelength shifts) and/or energy splittings (e.g., wavelength splittings) that are orders of magnitude smaller than those of a fine structure shift and/or split. For example, for the $5S_{1/2}$ ground state of ^{85}Rb , the hyperfine energy state is designated by the parameter F (e.g., $5S_{1/2}$, F=3).

Doppler broadening of spectral lines is a result of thermal motion of an atomic vapor at a finite temperature relative to an optical field. Due to the finite temperature, the atoms are thermally excited and different velocities of the emitting atoms cause different Doppler shifts (e.g., change in frequency of a wave in relation to an observer who is moving relative to the wave source). Different Doppler shifts in turn cause the cumulative effect of line broadening.

For example, electrons in Rb atoms are found in the ground state, $5S_{1/2}$, and upon excitation are moved to a higher energy state depending on the discrete energy received (e.g., 780.2 nm). The $5S_{1/2}$ to $5P_{3/2}$ atomic transition can be approximated as a two-level atomic system with a hyperfine ground state (e.g., $5S_{1/2}$, F=3 for ^{85}Rb , $5S_{1/2}$, F=2 for ^{87}Rb) and a Doppler-broadened excited state. Atoms whose electron has decayed into another (lower) hyperfine

ground state (e.g., $5S_{1/2}$, F=2 for ^{85}Rb , $5S_{1/2}$, F=1 for ^{87}Rb) can be treated as decohered atoms. Decohered atoms must be optically pumped back (increase in energy) into the ground state (e.g., $5S_{1/2}$, F=3 for ^{85}Rb , $5S_{1/2}$, F=2 for ^{87}Rb) before they can participate in the DFWM process.

FIG. 2 illustrates energy diagram 200 for DFWM, according to various exemplary embodiments. Energy diagram 200 can include a ground state 210 (e.g., $5S_{1/2}$, F=3), a Doppler-broadened excited state 220 (e.g., $5P_{3/2}$), and a virtual excited state 230 separated by a frequency detuning (Δ) 240 from Doppler-broadened excited state 220. Optical fields (e.g., pump beam(s) 202, probe beam 206) are frequency tuned to a frequency that is greater than the resonant ground-to-excited state 215 transition frequency by frequency detuning (Δ) 240. Resonant excitation (e.g., $\Delta=0$) causes nearly all optical fields to be absorbed and spontaneously reemitted and, thus, squeezing cannot occur when frequency detuning (Δ) 240 is zero (e.g., $\Delta=0$) or close to zero (e.g., $\Delta\approx 0$). In some embodiments, frequency detuning (Δ) 240 can be at the edge of or just outside Doppler-broadened excited state 220. For example, frequency detuning (Δ) 240 can be about 300 MHz to about 800 GHz.

As shown in FIG. 2, energy diagram 200 can approximate a two-level atomic system with pump beam(s) 202 driving atomic transitions between ground state 210 and virtual excited state 230 to amplify probe beam 206 and produce conjugate beam 208. Pump beam(s) 202 can drive first and second atomic transitions in an atomic vapor (e.g., Rb atoms) from ground state 210 to virtual excited state 230. In the first transition, pump beam 202 excites an atom (e.g., Rb atom) in the atomic vapor from ground state 210 to virtual excited state 230, and probe beam 206 induces stimulated emission from virtual excited state 230 to ground state 210. This excitation-emission process changes the dipole moment of the atomic vapor. In the second transition, pump beam 202 excites atoms (e.g., Rb atoms) in the atomic vapor with the new atomic dipole moment and includes emission that produces conjugate beam 208. The interaction of pump beam 202 with the dipole moment induced in the atoms by the second transition then, in a synergistic manner, amplifies probe beam 206 of the first transition. In some embodiments, ground state 210 can be F=3 for a ^{85}Rb vapor and/or F=2 for a ^{87}Rb vapor. For example, as shown in FIG. 2, ground state 210 can be F=3 for a ^{85}Rb vapor.

DFWM can utilize a two-level atomic system in which all beams have the same frequency. DFWM can occur even if only two beams interact (e.g., pump beam and probe beam). The pump and probe beams have the same frequency but different wavevectors (e.g., k-vectors) since they must propagate at a small angle relative to one another in order to conserve momentum in the DFWM process. The sum of the k-vectors of the pump beams must equal the sum of the k-vectors of the probe and conjugate beams.

In a backward-scattering DFWM geometry, two counter-propagating pump beams and a probe beam propagating at a small angle (e.g., small angle 305) relative to one of the pump beams overlap inside a nonlinear medium (e.g., an atomic vapor) and a conjugate beam is generated via a four-wave mixing mechanism. The forward pump beam and the probe beam propagate in one direction (e.g., forward) and the backward pump beam propagates in the opposite direction (e.g., backward). Thus, the conjugate beam (e.g., fourth beam) also comes out in the opposite direction (e.g., backward). The conjugate beam is a wavevector reversed replica of the probe beam (e.g., its phase is conjugate to that of the probe beam) and any classical noise present on the input probe beam will be canceled out upon joint detection.

Any sources of excess noise that arise after the nonlinear processes (e.g., that occur independently) on one or both beams will not be canceled out.

FIG. 3 illustrates backward-scattering geometry 300, according to various exemplary embodiments. Backward-scattering geometry 300 can include pump beams 302 and probe beam 304 impinging upon a vapor cell 310. Vapor cell 310 can include an atomic vapor 311 (e.g., Rb) enclosed within a container 312 extending a length 314. As shown in FIG. 3, the interaction of pump beams 302, probe beam 304, and atomic vapor 311 creates (e.g., via DFWM) amplified probe beam 306 and conjugate beam 308. Amplified probe beam 306 and conjugate beam 308 can be a squeezed light source, for example, two-mode squeezed light by DFWM with squeezing of at least 3 dB below shot noise. As shown in FIG. 3, pump beams 302 and probe beam 304 can propagate at small angle 305. In some embodiments, small angle 305 can be about 1 degree to about 10 degrees. For example, small angle 305 can be about 5 degrees. In some embodiments, backward-scattering geometry 300 can be utilized in DFWM squeezed light apparatus 500 and/or DFWM squeezed light system 900.

In a forward-scattering DFWM geometry, all input beams propagate in one direction (e.g., forward) and, thus, the conjugate beam (e.g., fourth beam) also propagates in the same direction (e.g., forward). All input beams can be focused and mixed using, for example, a single lens, which can create a high photon density and efficient wave mixing.

FIG. 4 illustrates forward-scattering geometry 400, according to various exemplary embodiments. Forward-scattering geometry 400 can include pump beam 402 and probe beam 404 impinging upon a vapor cell 410. Vapor cell 410 can include an atomic vapor 411 (e.g., Rb) enclosed within a container 412 extending a length 414. As shown in FIG. 4, the interaction of pump beam 402, probe beam 404, and atomic vapor 411 creates (e.g., via DFWM) amplified probe beam 406 and conjugate beam 408. Amplified probe beam 406 and conjugate beam 408 can be a squeezed light source, for example, two-mode squeezed light by DFWM with squeezing of at least 3 dB below shot noise. As shown in FIG. 4, pump and probe beams 402, 404 can propagate at small angle 405. In some embodiments, small angle 405 can be about 0.01 degrees to about 1 degree. For example, small angle 405 can be about 0.5 degrees. In some embodiments, forward-scattering geometry 400 can be utilized in DFWM squeezed light apparatus 500 and/or DFWM squeezed light system 900. For example, as shown in FIG. 5, DFWM squeezed light apparatus 500 can utilize a forward-scattering geometry similar to forward-scattering geometry 400 shown in FIG. 4.

FIGS. 5-7 illustrate DFWM squeezed light apparatus 500, according to various exemplary embodiments. DFWM squeezed light apparatuses and systems as discussed below can reduce excess noise and provide a low-power and portable squeezed light source with a reduced size, weight, and power (SWaP) to improve the precision of optical measurements and/or enable continuous-variable quantum communication protocols.

FIG. 5 illustrates DFWM squeezed light apparatus 500, according to various exemplary embodiments. DFWM squeezed light apparatus 500 can be configured to generate squeezed light with an input power of no greater than about 150 mW. DFWM squeezed light apparatus 500 can be configured to generate two-mode squeezed light by DFWM with squeezing of at least 3 dB below shot noise. Although DFWM squeezed light apparatus 500 is shown in FIG. 5 as a stand-alone apparatus and/or system, the embodiments of

this disclosure can be used with other optical systems, such as, but not limited to, a portable diode laser, a portable integrated photonic chip, DFWM squeezed light system 900, and/or other optical systems.

DFWM squeezed light apparatus 500 can include pump beam 502, probe beam 504, amplified probe beam 506, conjugate beam 508, vapor cell 510, heating system 520, temperature sensing system 530, processor 540, optical block 542, balanced differential detector 550, and/or repump beam 590. Similar to forward-scattering geometry 400 shown in FIG. 4, pump beam 502 and probe beam 504 can propagate at a small angle 505 and overlap in vapor cell 510 to generate amplified probe beam 506 and conjugate beam 508 by DFWM. Pump beam 502, probe beam 504, and vapor cell 510 can be configured to generate two-mode squeezed light (e.g., amplified probe beam 506 and conjugate beam 508) by DFWM, for example, with squeezing of at least 3 dB below shot noise. In some embodiments, DFWM squeezed light apparatus 500 can generate two-mode squeezed light by DFWM with squeezing of at least 6 dB below shot noise. In some embodiments, DFWM squeezed light apparatus 500 can generate single-mode squeezed light by DFWM. In some embodiments, pump beam 502, probe beam 506, and/or repump beam 590 can be produced by an optical source (not shown). For example, as shown in FIG. 10, optical source 932 can be a compact coherent light source (e.g., diode laser, distributed feedback (DFB) laser, etc.) configured to produce pump beam 502, probe beam 506, and/or repump beam 590.

Pump beam 502 can be configured to excite an atomic vapor 511 in vapor cell 510, for example, from a ground state (e.g., ground state 210, e.g., $5S_{1/2}$, shown in FIG. 2) to an excited state (e.g., virtual excited state 230 shown in FIG. 2). Pump beam 502 can include an input power of no greater than about 150 mW. In some embodiments, the input power of pump beam 502 can be no greater than about 50 mW. In some embodiments, the input power of pump beam 502 can be no greater than about 20 mW. For example, pump beam 502 can be produced by a low-power portable diode laser and/or a DFB laser. In some embodiments, pump beam 502 can be produced by a pump source (not shown). For example, as shown in FIG. 10, optical source 932 can be a compact coherent light source (e.g., diode laser, DFB laser, etc.) configured to produce pump beam 502. In some embodiments, pump beam 502 can be polarized (e.g., linearly, circularly, etc.). For example, pump and probe beams 502, 504 can be cross-polarized (e.g., orthogonal). In some embodiments, pump beam 502 can include a wavelength in the visible, near-infrared, and/or infrared. For example, the wavelength of pump beam 502 can be about 780.2 nm, for example, when atomic vapor 511 comprises Rb.

Probe beam 504 can be configured to overlap pump beam 502 inside vapor cell 510 such that pump beam 502, probe beam 504, and atomic vapor 511 can undergo DFWM. Probe beam 504 can include an input power of no greater than about 15 mW (e.g., an order of magnitude lower than pump beam 502). In some embodiments, the input power of probe beam 504 is about 1 μ W to about 0.5 mW. For example, probe beam 504 can be about 100 μ W. In some embodiments, probe beam 504 can include a portion of pump beam 502. For example, a portion of pump beam 502 can be siphoned off via a beamsplitter (e.g., 99:1) to form separate probe beam 504. In some embodiments, probe beam 504 can be produced by a probe source (not shown) and/or a pump source (not shown). For example, as shown in FIG. 10, optical source 932 can be a compact coherent light source (e.g., diode laser, DFB laser, etc.) configured to produce

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probe beam **504**. In some embodiments, probe beam **504** can be polarized (e.g., linearly, circularly, etc.). For example, pump and probe beams **502**, **504** can be cross-polarized (e.g., orthogonal). In some embodiments, probe beam **504** can include a wavelength in the visible, near-infrared, and/or infrared. For example, the wavelength of probe beam **504** can be about 780.2 nm, for example, when atomic vapor **511** comprises Rb. As shown in FIG. 5, pump and probe beams **502**, **504** can propagate at small angle **505**. In some embodiments, small angle **505** can be about 0.01 degrees to about 1 degrees. For example, small angle **505** can be about 0.5 degrees.

Vapor cell **510** can be configured to enclose an atomic vapor **511**, for example, vaporized Rb (e.g., ⁸⁵Rb and/or ⁸⁷Rb). As shown in FIG. 5, vapor cell **510** can include atomic vapor **511**, cylindrical wall **512**, first window **516**, second window **518**, heating system **520**, temperature sensing system **530**, processor **540**, and/or balanced differential detector **550**. Cylindrical wall **512** can include exterior surface **513** and interior surface **515**. In some embodiments, as shown in FIG. 5, pump beam **502**, probe beam **504**, and vapor cell **510** can be arranged in a forward-scattering geometry (e.g., forward-scattering geometry **400** shown in FIG. 4). In some embodiments, as shown in FIG. 5, vapor cell **510** can have a longitudinal length **514**. In some embodiments, longitudinal length **514** can be about 0.5 cm to about 10 cm. For example, longitudinal length **514** can be about 8 cm. In some embodiments, vapor cell **510** can have a temperature (e.g., internal) of about 30° C. to about 100° C. For example, the temperature can be about 35° C. to about 45° C.

Atomic vapor **511** can be configured to interact with overlapped pump and probe beams **502**, **504** to generate (e.g., via DFWM) amplified probe beam **506** and conjugate beam **508**. In some embodiments, atomic vapor **511** can include a rubidium (Rb) vapor. In some embodiments, atomic vapor **511** can include an alkali metal (e.g., sodium (Na), rubidium (Rb), caesium (Cs), etc.). Amplified probe beam **506** can be produced via DFWM and can have an output power of about 1 μW to about 15 mW. For example, probe beam **504** can be about 100 μW and amplified probe beam **506** can be about 1 mW, which corresponds to a gain of 10, where $\text{gain} = (\text{output power of amplified probe beam } 506) / (\text{input power of probe beam } 504)$. Conjugate beam **508** can be produced via DFWM and can have an output power of about 1 μW to about 15 mW. Output power of conjugate beam **508** can be approximately equal to $(\text{input probe beam } 504) * (\text{gain} - 1)$.

As shown in FIG. 5, cylindrical wall **512** can be disposed between first and second windows **516**, **518** to enclose (e.g., seal under vacuum) atomic vapor **511**. In some embodiments, cylindrical wall **512** can include copper, glass, and/or any other material with a high thermal conductivity. First and second windows **516**, **518** can be configured to transmit (e.g., focus) pump and probe beams **502**, **504** into atomic vapor **511** and transmit generated amplified probe beam **506** and conjugate beam **508** to balanced differential detector **550**. In some embodiments, as shown in FIG. 5, first and second windows **516**, **518** can be disks. In some embodiments, first and second windows **516**, **518** can include an anti-reflection (AR) coating. For example, the AR coating can be a selective wavelength notch filter, for example, a bandpass filter of about 780.2 nm. In some embodiments, a focusing element and/or a lens (not shown) can be disposed near and/or integrated with first window **516** and can be configured to focus pump and probe beams **502**, **504** into atomic vapor **511**.

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Heating system **520** can be configured to increase and/or decrease a temperature of vapor cell **510** and/or atomic vapor **511**. Heating system **520** can include first and second heaters **522**, **524** (e.g., resistive coils) configured to provide heating (e.g., resistive). Heating system **520** can be coupled (e.g., electrically) to vapor cell **510**, for example, first and second windows **516**, **518**. In some embodiments, first and second windows **516**, **518** can have a temperature (e.g., exterior surface temperature) greater than an exterior surface **513** of cylindrical wall **512**. For example, as shown in FIG. 5, first and second windows **516**, **518** can include first and second heaters **522**, **524** (e.g., resistive coils) disposed radially around first and second windows **516**, **518**, respectively. In some embodiments, first and second heaters **522**, **524** (e.g., resistive coils) can be configured to heat first and second windows **516**, **518** such that a thermal gradient is formed inside vapor cell **510**. For example, the thermal gradient can have a cooler region about a centroid of cylindrical wall **512** to condense atomic vapor **511** and reduce adsorption, absorption, and/or Van der Waals interactions of atomic vapor **511** on first and second windows **516**, **518**. In some embodiments, heating system **520** can provide vapor cell **510** and/or atomic vapor **511** with a temperature (e.g., internal) of about 30° C. to about 100° C. For example, the temperature of vapor cell **510** and/or atomic vapor **511** can be about 35° C. to about 45° C. In some embodiments, first and second heaters **522**, **524** can be resistive, electric, thermoelectric, radiative, convective, heat pump, Peltier, etc. For example, as shown in FIG. 5, first and second heaters **522**, **524** can be resistive coils.

Temperature sensing system **530** can be configured to measure and/or monitor a temperature of vapor cell **510** and/or atomic vapor **511**. Temperature sensing system **530** can include first and second temperature sensors **532**, **534** (e.g., thermocouples) configured to sense a temperature of vapor cell **510**, for example, exterior surface **513** of cylindrical wall **512**. Temperature sensing system **530** can be coupled (e.g., electrically) to vapor cell **510**, for example, exterior surface **513** of cylindrical wall **512**. In some embodiments, temperature sensing system **530** can provide a feedback signal to processor **540** and/or heating system **520** in order to maintain, increase, and/or decrease a temperature of vapor cell **510** and/or atomic vapor **511**. In some embodiments, first and second temperature sensors **532**, **534** can be a thermocouple, a thermistor, a resistance temperature detector (RTD), an infrared sensor, etc. For example, as shown in FIG. 5, first and second heaters **522**, **524** can be thermocouples disposed on exterior surface **513** of cylindrical wall **512**.

Processor **540** can be configured to control heating system **520**, temperature sensing system **530**, and/or balanced differential detector **550**. For example, processor **540** can receive temperature signals from temperature sensing system **530** and adjust heating system **520** (e.g., provide voltage and/or current) to maintain a specific temperature of vapor cell **510** and/or atomic vapor **511**. Further, processor **540** can measure squeezing between amplified probe beam **506** and conjugate beam **508**, for example, squeezing of at least 3 dB below shot noise. As shown in FIG. 5, processor **540** can be coupled (e.g., electrically) to heating system **520**, temperature sensing system **530**, and/or balanced differential detector **550**. In some embodiments, processor **540** can maintain a temperature on first and second windows **516**, **518** greater than exterior surface **513** of cylindrical wall **512**. In some embodiments, processor **540** can maintain vapor cell **510** and/or atomic vapor **511** at a temperature (e.g., internal) of

about 30° C. to about 100° C. For example, the temperature of vapor cell **510** and/or atomic vapor **511** can be about 35° C. to about 45° C.

Optical block **542** can be configured to utilize two-mode squeezed light (e.g., amplified probe beam **206** and conjugate beam **208**) produced by DFWM squeezed light apparatus **500**. For example, optical block **542** can conduct optical measurements (e.g., user experiment(s)) and/or implement continuous-variable quantum communication protocols (e.g., encryption of information along a communication channel) with the two-mode squeezed light (e.g., amplified probe beam **206** and conjugate beam **208**) prior to amplified probe beam **206** and conjugate beam **208** being detected. In some embodiments, as shown in FIG. 5, optical block **542** can be disposed between the outputted two-mode squeezed light (e.g., amplified probe beam **206** and conjugate beam **208**) and balanced differential detector **550**. In some embodiments, optical block **542** can be omitted and is an optional optical path of amplified probe beam **206** and conjugate beam **208**.

Balanced differential detector **550** can be configured to measure squeezing between amplified probe beam **506** and conjugate beam **508**. As shown in FIG. 5, balanced differential detector **550** can include intensity-difference detector **560** with first and second detectors **562**, **564** and intensity-difference output **566**. First and second detectors **562**, **564** can be configured to receive amplified probe beam **506** and conjugate beam **508**, respectively, and intensity-difference output **566** can be coupled to processor **540**. Intensity-difference detector **560** can be configured to subtract amplified probe beam **506** and conjugate beam **508** intensity signals measured by first and second detectors **562**, **564**, respectively. In some embodiments, balanced differential detector **550** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 3 dB below shot noise. In some embodiments, balanced differential detector **550** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 6 dB below shot noise. For example, as shown in FIG. 11, squeezing plot **1100** (e.g., for intensity-difference detection, forward-scattering geometry, as shown in FIG. 5) for about zero decoherence (e.g., decoherence rate $\Gamma_{23}=0$) indicates that DFWM squeezed light apparatus **500** can achieve (e.g., an input power density of 10 W/cm² for pump beam **502**) squeezing of about 7 dB below shot noise for a quantum efficiency (η) of 0.8 for balanced differential detector **550**. In some embodiments, balanced differential detector **550** and/or processor **540** can include a spectrum analyzer (not shown).

In some embodiments, DFWM squeezed light apparatus **500** can include balanced differential detector **550** and/or joint homodyne detector **550'**. For example, as shown in FIG. 6, DFWM squeezed light apparatus **500** can include joint homodyne detector **550'**. In some embodiments, balanced differential detector **550** can include an intensity-difference detector **560**. In some embodiments, as shown in FIG. 6, joint homodyne detector **550'** can include homodyne detectors **560a**, **560b**.

In some embodiments, DFWM squeezed light apparatus **500** can include a portable diode laser (not shown) configured to provide pump beam **502**, probe beam **504**, and/or repump beam **590**. For example, as shown in FIG. 10, DFWM squeezed light apparatus **500** can include optical source **932** that can be a portable diode laser. In some embodiments, DFWM squeezed light apparatus **500** can include a portable diode laser (not shown) that can have a volume of no greater than about 100 cm³. In some embodi-

ments, DFWM squeezed light apparatus **500** can include a portable diode laser (not shown) that can have an output power of no greater than about 150 mW. In some embodiments, DFWM squeezed light apparatus **500** can include a portable integrated photonic chip (not shown). For example, as shown in FIGS. 9 and 10, DFWM squeezed light apparatus **500** can be incorporated in DFWM squeezed light system **900** that can include a portable integrated photonic chip.

In some embodiments, DFWM squeezed light apparatus **500** can include repump beam **590** configured to optically pump atomic vapor **511** to a ground state (e.g., ground state **210**, e.g., $5S_{1/2}$, $F=3$ shown in FIG. 2) and decrease the atomic decoherence rate (Γ_{23}) of atomic vapor **511**. For example, as shown in FIGS. 7, 7A, and 8, repump beam **590** of DFWM squeezed light apparatus **500** can be utilized to optically pump atomic vapor **511**, decrease the atomic decoherence rate (Γ_{23} , written in FIG. 11 in terms of the spontaneous emission rate Γ), and lower noise (e.g., in decibel-milliwatts) of DFWM squeezed light apparatus **500**. In some embodiments, the spontaneous emission rate Γ is approximately $2\pi \times 6$ MHz, for example, for Rb.

FIG. 6 illustrates another exemplary detector, joint homodyne detector **550'**, for DFWM squeezed light apparatus **500** shown in FIG. 5, according to an exemplary embodiment. The embodiments of balanced differential detector **550** shown in FIG. 5 and the embodiments of joint homodyne detector **550'** shown in FIG. 6 may be similar. Similar reference numbers are used to indicate similar features of the embodiments of balanced differential detector **550** shown in FIG. 5 and the similar features of the embodiments of joint homodyne detector **550'** shown in FIG. 6. The main difference between the embodiments of balanced differential detector **550** shown in FIG. 5 and the embodiments of joint homodyne detector **550'** shown in FIG. 6 is that joint homodyne detector **550'** includes homodyne detectors **560a**, **560b** configured to detect quadrature-squeezed noise of DFWM squeezed light apparatus **500** from amplified probe beam **506** and conjugate beam **508**, rather than intensity-difference squeezed noise from intensity-difference detector **560** shown in FIG. 5. Although joint homodyne detector **550'** is shown in FIG. 6 as a stand-alone apparatus and/or system, the embodiments of this disclosure can be used with other optical systems, such as, but not limited to, DFWM squeezed light apparatus **500**, DFWM squeezed light system **900**, and/or other optical systems.

As shown in FIG. 6, an exemplary aspect of joint homodyne detector **550'** is homodyne detectors **560a**, **560b**. Joint homodyne detector **550'** can be configured to measure squeezing between amplified probe beam **506** and conjugate beam **508**. As shown in FIG. 6, joint homodyne detector **550'** can include homodyne detectors **560a**, **560b**. Homodyne detection uses a single reference frequency to extract a modulated signal from a noisy background. For example, homodyne detection extracts encoded information from an oscillating signal (e.g., phase and/or frequency) by comparing that signal with a standard reference oscillation (e.g., identical to the signal if it carried null information).

First homodyne detector **560a** can be configured to measure amplified probe beam **506**. First homodyne detector **560a** can include first and second detectors **562a**, **564a**, first homodyne output **566a**, first phase shifting element **570a** (e.g., acousto-optic modulator (AOM), electro-optic modulator (EOM), piezoelectric (PZT) transducer, etc.), first reference beam **572a** (e.g., first phase shifting element **570a** can be configured to measure a particular quadrature of amplified probe beam **506**), first beamsplitter **574a**, and/or

first mirror **576a**. First beamsplitter **574a** can be configured to combine amplified probe beam **506** and first reference beam **572a** and direct them to first mirror **576a** and second detector **564a**. First mirror **576a** can be configured to direct overlapped amplified probe beam **506** and first reference beam **572a** to first detector **562a**. In some embodiments, first mirror **576a** can be omitted and overlapped amplified probe beam **506** and first reference beam **572a** can be coupled directly into first detector **562a**. First and second detectors **562a**, **564a** can be configured to receive first and second components of overlapped amplified probe beam **506** and first reference beam **572a** (e.g., formed by 50:50 beamsplitter), respectively, and first homodyne output **566a** can be coupled to second homodyne output **566b** and/or processor **540**. First homodyne detector **560a** can be configured to subtract first and second components of amplified probe beam **506** intensity signals measured by first and second detectors **562a**, **564a**, respectively, and extract noise quadrature statistics of amplified probe beam **506** relative to first reference beam **572a**.

Second homodyne detector **560b** can be configured to measure conjugate beam **508**. Similarly, second homodyne detector **560b** can include first and second detectors **562b**, **564b**, second homodyne output **566b**, second phase shifting element **570b** (e.g., acousto-optic modulator (AOM), electro-optic modulator (EOM), piezoelectric (PZT) transducer, etc.), second reference beam **572b** (e.g., second phase shifting element **570b** can be configured to measure a particular quadrature of conjugate beam **508**), second beamsplitter **574b**, and second mirror **576b**. Second beamsplitter **574b** can be configured to combine conjugate beam **508** and second reference beam **572b** and direct them to second mirror **576b** and second detector **564b**. Second mirror **576b** can be configured to direct overlapped conjugate beam **508** and second reference beam **572b** to first detector **562b**. In some embodiments, second mirror **576b** can be omitted and overlapped conjugate beam **508** and second reference beam **572b** can be coupled directly into first detector **562b**. First and second detectors **562b**, **564b** can be configured to receive first and second components of overlapped conjugate beam **508** and second reference beam **572b** (e.g., formed by 50:50 beamsplitter), respectively, and second homodyne output **566b** can be coupled to first homodyne output **566a** and/or processor **540**. Second homodyne detector **560b** can be configured to subtract first and second components of conjugate beam **508** intensity signals measured by first and second detectors **562b**, **564b**, respectively, and extract noise quadrature statistics of conjugate beam **508** relative to second reference beam **572b**.

In some embodiments, joint homodyne detector **550** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 3 dB below shot noise. In some embodiments, joint homodyne detector **550** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 6 dB below shot noise.

FIGS. 7 and 7A illustrate repump beam **590** in DFWM squeezed light apparatus **500** shown in FIG. 5, according to an exemplary embodiment. Repump beam **590** can be configured to reduce noise in DFWM squeezed light apparatus **500**. Repump beam **590** can be configured to optically pump atomic vapor **511** to a ground state (e.g., ground state **210**, e.g., $5S_{1/2}$, $F=3$ shown in FIG. 2) and decrease the atomic decoherence rate (Γ_{23}) of atomic vapor **511**. For example, repump beam **590** can be configured to reduce spontaneous emission into the spatial modes of amplified probe beam **506** and conjugate beam **508**. Repump beam **590** can cause

spontaneous emission of atoms in atomic vapor **511** near the edges (e.g., interior surface **515**) of vapor cell **510**, and can reduce spontaneous emission of atoms in atomic vapor **511** involved in the DFWM process (e.g., near center of vapor cell **510**). In some embodiments, DFWM squeezed light apparatus **500** can include one or more repump beams **590** configured to optically pump atomic vapor **511** to a ground state (e.g., ground state **210**, e.g., $5S_{1/2}$, $F=3$ shown in FIG. 2) and decrease the atomic decoherence rate (e.g., Γ_{23} shown in FIG. 11) of atomic vapor **511**. For example, DFWM squeezed light apparatus **500** can be arranged in a backward-scattering geometry (e.g., backward-scattering geometry **300** shown in FIG. 3) and two counter-propagating collimated repump beams (e.g., each similar to repump beam **590** shown in FIG. 5) can interact with atomic vapor **511** of DFWM squeezed light apparatus **500**. In some embodiments, repump beam **590** can be from the same optical source (e.g., portable diode laser, etc.) as pump and probe beams **502**, **504**, for example, optical source **932** shown in FIG. 10.

FIG. 7A is a partial cross-sectional view of repump beam **590**. In some embodiments, repump beam **590** can be cylindrical, for example, a solid cylinder with a solid elliptical cross-section along an optical axis of repump beam **590**. For example, as shown in FIGS. 7 and 7A, repump beam **590** can be an annulus **592** (e.g., a ring with a hollow center), for example, with an annular cross-section along an optical axis of repump beam **590**. In some embodiments, repump beam **590** can include two counter-propagating annular repump beams (e.g., similar to repump beam **590** with annulus **592**), for example, in a backward-scattering geometry (e.g., backward-scattering geometry **300** as shown in FIG. 3). The annular shape **592** of repump beam **590** can be configured to decrease the atomic decoherence rate (Γ_{23}) of atomic vapor **511** by promoting optical pumping of atomic vapor **511**, particularly for those atoms of atomic vapor **511** near (e.g., via thermal drifting) interior surface **515** that have undergone spin-changing collisions with of cylindrical wall **512**. As shown in FIG. 8, noise plot **800** with noise signal without repump beam **802** and noise signal with repump beam **804** indicates that noise (e.g., in decibel-milliwatts (dBm)) of DFWM squeezed light apparatus **500** can be lowered with the addition of repump beam **590** in vapor cell **510**. In some embodiments, repump beam **590** can be collimated through vapor cell **510** (e.g., not focused into atomic vapor **511** by any focusing element).

In some embodiments, repump beam **590** can have an input power similar to that of an input power of pump beam **502**. For example, the input power of repump beam **590** can be approximately the same as an input power of pump beam **502** (e.g., about 20 mW to about 150 mW). In some embodiments, repump beam **590** can have an input power no greater than about an input power of pump beam **502**. For example, the input power of repump beam **590** can be no greater than about 150 mW. In some embodiments, pump beam **502**, probe beam **504**, and repump beam **590** can be configured to reduce spontaneous emission of atomic vapor **511** and generate two-mode squeezed light by DFWM in atomic vapor **511**.

In some embodiments, DFWM squeezed light apparatus **500** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 3 dB below shot noise. In some embodiments, DFWM squeezed light apparatus **500** can measure squeezing of amplified probe beam **506** and conjugate beam **508** by DFWM of at least 6 dB below shot noise. For example, as shown in FIG. 11, squeezing plot **1100** (e.g., for intensity-difference detection,

forward-scattering geometry, as shown in FIG. 5) for about zero decoherence (e.g., decoherence rate $\Gamma_{23}=0$) indicates that DFWM squeezed light apparatus 500 can achieve (e.g., an input power density of 10 W/cm² for pump beam 502) squeezing of about 7 dB below shot noise for a quantum efficiency (η) of 0.8 for balanced differential detector 550. Exemplary DFWM Squeezed Light System

FIGS. 9 and 10 illustrate DFWM squeezed light system 900, according to various exemplary embodiments. DFWM squeezed light system 900 can reduce excess noise and provide a low-power and portable squeezed light source with a reduced size, weight, and power (SWaP) to improve precision of optical measurements and/or enable certain quantum communication protocols.

FIGS. 9 and 10 illustrate DFWM squeezed light system 900, according to an exemplary embodiment. The embodiments of DFWM squeezed light apparatus 500 shown in FIGS. 5-7 and the embodiments of DFWM squeezed light system 900 shown in FIGS. 9 and 10 may be similar. Similar reference numbers are used to indicate similar features of the embodiments of DFWM squeezed light apparatus 500 shown in FIGS. 5-7 and the similar features of the embodiments of DFWM squeezed light system 900 shown in FIGS. 9 and 10. One difference between the embodiments of DFWM squeezed light apparatus 500 shown in FIGS. 5-7 and the embodiments of DFWM squeezed light system 900 shown in FIGS. 9 and 10 is that DFWM squeezed light system 900 includes portable integrated photonic chip 910 and optical input system 930 to provide a low-power and portable optical system including DFWM squeezed light apparatus 500, for example, with a portable integrated photonic chip (e.g., portable integrated photonic chip 910) and a portable diode laser (e.g., optical input system 930). Although DFWM squeezed light system 900 is shown in FIGS. 9 and 10 as a stand-alone apparatus and/or system, the embodiments of this disclosure can be used with other optical systems, such as, but not limited to, DFWM squeezed light apparatus 500 and/or other optical systems.

DFWM squeezed light system 900 can be configured to provide a low-power and portable squeezed light source. DFWM squeezed light system 900 can include portable integrated photonic chip 910, optical input system 930, and DFWM squeezed light apparatus 500 with atomic vapor 511 enclosed by first and second windows 516, 518 and cylindrical wall 512. As shown in FIGS. 9 and 10, DFWM squeezed light apparatus 500 can be disposed between portable integrated photonic chip 910 and optical input system 930. In some embodiments, DFWM squeezed light system 900 can include DFWM squeezed light apparatus 500. In some embodiments, as shown in FIGS. 9 and 10, DFWM squeezed light apparatus 500 can be arranged in a forward-scattering geometry (e.g., forward-scattering geometry 400 as shown in FIG. 4) with optical input system 930 providing pump beam 502, probe beam 504, and repump beam 590 and portable integrated photonic chip 910 receiving amplified probe beam 506 and conjugate beam 508 and coupling amplified probe beam 506 and conjugate beam 508 to optical block 542, balanced differential detector 550, and/or joint homodyne detector 550'. In some embodiments, DFWM squeezed light apparatus 500 can be arranged in a backward-scattering geometry (e.g., backward-scattering geometry 300 as shown in FIG. 3) with optical input system 930 providing pump beams 502, probe beam 504, and repump beams 590 (e.g., secondary pump and repump beams 502, 590 can be routed to second window 518 side via optics and/or additional waveguides) and portable integrated photonic chip 910 receiving amplified probe beam 506 and

conjugate beam 508 (e.g., conjugate beam 508 can be routed to second window 518 side via optics and/or additional waveguides) and coupling amplified probe beam 506 and conjugate beam 508 to optical block 542, balanced differential detector 550, and/or joint homodyne detector 550'.

As shown in FIGS. 9 and 10, an exemplary aspect of DFWM squeezed light system 900 is portable integrated photonic chip 910. Portable integrated photonic chip 910 can be configured to output amplified probe beam 506 and conjugate beam 508 from DFWM squeezed light apparatus 500 to optical block 542 (e.g., user experiment(s), optical measurement(s), communication encryption, quantum communication protocols, etc.), balanced differential detector 550, and/or joint homodyne detector 550'. In some embodiments, as shown in FIG. 10, optical block 542 can be disposed between the outputted two-mode squeezed light (e.g., amplified probe beam 206 and conjugate beam 208) from optical fiber(s) 908 and balanced differential detector 550 and/or joint homodyne detector 550'. In some embodiments, optical block 542 can be omitted and is an optional optical path of amplified probe beam 206 and conjugate beam 208 (e.g., via optical fiber(s) 908). Portable integrated photonic chip 910 can include substrate 902, intermediate layer 904 (e.g., oxide and/or nitride), waveguide 912 with input/output grating 914, waveguide coupling 916 with etched tip 918, and optical coupling 906 with optical fiber(s) 908. As shown in FIG. 10, amplified probe beam 506 and conjugate beam 508 can optically couple to input/output grating 914 of waveguide 912. For example, second window 518 can be disposed atop input/output grating 914. Waveguide coupling 916 can optically couple to optical fiber(s) 908, for example, via etched tip 918. As shown in FIG. 10, optical coupling 906 can be optically coupled (e.g., via optical fiber(s) 908) to optical block 542, balanced differential detector 550, and/or joint homodyne detector 550' in order to utilize and/or measure squeezing between amplified probe beam 506 and conjugate beam 508.

As shown in FIGS. 9 and 10, an exemplary aspect of DFWM squeezed light system 900 is optical input system 930. Optical input system 930 can be configured to input and/or focus pump beam 502, probe beam 504, and/or repump beam 590 into DFWM squeezed light apparatus 500. Optical input system 930 can include optical source 932, beam routing optics 934, and/or focusing optics 920. Optical source 932 can be configured to provide pump beam(s) 502, probe beam 504, and/or repump beam(s) 590. Beam routing optics 934 can be configured to separate pump beam(s) 502, probe beam 504, and/or repump beam(s) 590 and route them to appropriate positions before being projected into DFWM squeezed light apparatus 500 (e.g., forward-scattering geometry, backward-scattering geometry). For example, repump beam 590 can be collimated and beam expanded to encompass pump and probe beams 502, 504 that are arranged relative to each other by small angle 505, and pump and probe beams 502, 504 can be routed toward focusing optics 920 while repump beam 590 bypasses focusing optics 920. Focusing optics 920 can be configured to focus pump beam 502 and probe beam 504 into atomic vapor 511. Repump beam 590 can be collimated through DFWM squeezed light apparatus 500 (e.g., as shown in FIGS. 5, 7, and 10).

In some embodiments, optical source 932 can provide pump beam 502, probe beam 504, and/or repump beam 590. In some embodiments, optical sources for pump beam 502, probe beam 504, and/or repump beam 590 can be combined into a single optical power source and optic assembly (e.g., optical source 932), for example, a portable diode laser. In

some embodiments, optical source **932**, beam routing optics **934**, and/or focusing optics **920** can be combined into a single compact optical assembly (e.g., optical input system **930**). In some embodiments, optical source **932** can include a portable diode laser. In some embodiments, DFWM squeezed light system **900** can have a volume of no greater than about 100 cm³. In some embodiments, optical input system **930** can have a volume of no greater than about 100 cm³. In some embodiments, optical input system **930** can have an output power of no greater than about 150 mW. For example, optical source **932** can have an output power of no greater than about 150 mW.

Exemplary Flow Diagram

FIG. **12** illustrates flow diagram **1200** for forming a low-power squeezed light source, according to an embodiment. It is to be appreciated that not all steps in FIG. **12** are needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, sequentially, or in a different order than shown in FIG. **12**. Flow diagram **1200** shall be described with reference to FIGS. **5-7**. However, flow diagram **1200** is not limited to those example embodiments.

In step **1202**, as shown in the example of FIG. **5**, pump beam **502** and probe beam **504** can overlap in vapor cell **510**. In some embodiments, pump beam **502** can include an input power of no greater than about 150 mW. In some embodiments, pump beam **502** can include an input power of no greater than about 50 mW.

In step **1204**, as shown in the example of FIG. **5**, overlapped pump and probe beams **502**, **504** can interact with atomic vapor **511** in vapor cell **510** to generate amplified probe beam **506** and conjugate beam **508**.

In step **1206**, as shown in the examples of FIGS. **5-7**, repump beam **590** can optically pump atomic vapor **511** to decrease atomic decoherence of atomic vapor **511**. In some embodiments, repump beam **590** can include an annular repump beam **592**.

In step **1208**, as shown in the examples of FIGS. **5-7**, two-mode squeezed light can be generated by degenerate four-wave mixing (DFWM) in atomic vapor **511**. In some embodiments, two-mode squeezed light can be generated with squeezing of at least 3 dB below shot noise. In some embodiments, two-mode squeezed light can be generated with squeezing of at least 6 dB below shot noise.

In some embodiments, flow diagram **1200** can further include measuring an intensity-difference squeezing (e.g., with intensity-difference detector **560**) between amplified probe beam **506** and conjugate beam **508** with balanced differential detector **550**. For example, as shown in FIG. **11**, squeezing plot **1100** (e.g., for intensity-difference detection, forward-scattering geometry, as shown in FIG. **5**) for about zero decoherence (e.g., decoherence rate $\Gamma_{23}=0$) indicates that DFWM squeezed light apparatus **500** can achieve (e.g., an input power density of 10 W/cm² for pump beam **502**) squeezing of about 7 dB below shot noise for a quantum efficiency (η) of 0.8 for balanced differential detector **550**. In some embodiments, flow diagram **1200** can further include measuring quadrature squeezing (e.g., homodyne detectors **560a**, **560b**) between amplified probe beam **506** and conjugate beam **508** with joint homodyne detector **550**.

It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in the relevant art(s) in light of the teachings herein.

The above examples are illustrative, but not limiting, of the embodiments of this disclosure. Other suitable modifi-

cations and adaptations of the variety of conditions and parameters normally in the field, and which would be apparent to those skilled in the relevant art(s), are within the spirit and scope of the disclosure.

While specific embodiments have been described above, it will be appreciated that the embodiments may be practiced otherwise than as described. The description is not intended to limit the scope of the claims.

It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit the embodiments and the appended claims in any way.

The embodiments have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments that others can, by applying knowledge within the skill of the art(s), readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the embodiments. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein.

The breadth and scope of the embodiments should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

What is claimed is:

1. A degenerate four-wave mixing (DFWM) squeezed light apparatus comprising:

- a pump beam;
 - a probe beam configured to overlap the pump beam;
 - a vapor cell comprising an atomic vapor, wherein the atomic vapor is configured to interact with the overlapped pump and probe beams to generate an amplified probe beam and a conjugate beam;
 - a repump beam configured to optically pump the atomic vapor to a ground state; and
 - a detector configured to measure squeezing due to quantum correlations between the amplified probe beam and the conjugate beam,
- wherein the pump beam and the probe beam are configured to generate two-mode squeezed light by DFWM in the atomic vapor.

2. The DFWM squeezed light apparatus of claim 1, wherein the repump beam is configured to decrease atomic decoherence of the atomic vapor.

3. The DFWM squeezed light apparatus of claim 1, wherein the repump beam is configured to reduce noise of the DFWM squeezed light apparatus.

4. The DFWM squeezed light apparatus of claim 1, wherein the repump beam is annular.

5. The DFWM squeezed light apparatus of claim 1, wherein an input power of the repump beam is no greater than an input power of the pump beam.

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6. The DFWM squeezed light apparatus of claim 1, wherein the repump beam comprises two counter-propagating repump beams.

7. The DFWM squeezed light apparatus of claim 1, wherein the two-mode squeezed light is configured for continuous-variable entanglement.

8. The DFWM squeezed light apparatus of claim 1, wherein the two-mode squeezed light is configured for continuous-variable quantum communication protocols.

9. The DFWM squeezed light apparatus of claim 1, wherein the two-mode squeezed light comprises a continuous spectrum.

10. The DFWM squeezed light apparatus of claim 1, wherein the two-mode squeezed light comprises squeezing of at least 3 dB below shot noise.

11. A method comprising:

overlapping a pump beam and a probe beam in a vapor cell comprising an atomic vapor;

interacting the atomic vapor with the overlapped pump and probe beams;

optically pumping the atomic vapor with a repump beam; and

generating two-mode squeezed light by degenerate four-wave mixing (DFWM) in the atomic vapor.

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12. The method of claim 11, wherein optically pumping comprises decreasing atomic decoherence of the atomic vapor.

13. The method of claim 11, wherein optically pumping comprises reducing noise of the two-mode squeezed light.

14. The method of claim 11, wherein optically pumping comprises applying an annular repump beam.

15. The method of claim 11, wherein optically pumping comprises applying an input power of the repump beam that is no greater than an input power of the pump beam.

16. The method of claim 11, wherein optically pumping comprises applying two counter-propagating repump beams.

17. The method of claim 11, further comprising utilizing the two-mode squeezed light for continuous-variable entanglement.

18. The method of claim 11, further comprising utilizing the two-mode squeezed light for continuous-variable quantum communication protocols.

19. The method of claim 11, wherein interacting comprises generating an amplified probe beam and a conjugate beam.

20. The method of claim 19, further comprising measuring squeezing due to quantum correlations between the amplified probe beam and the conjugate beam.

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